CCAR-F FDE Track B (Accelerated)

Bonus / Standalone References (outside the 70-chapter core track) — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: docs/<slug>/SUMMARY.md + README.md + quiz.md (self-authored fixture, not a live book <Quiz> component).

EN: Panaversity's 13-week accelerated syllabus with two strands: Architect (CCAR-F exam prep, 70% grade) and FDE Practicum (build a governed Vertical System of Record, 30% grade).
Roman Urdu: Panaversity ka 13-week accelerated syllabus, do strands mein: Architect (CCAR-F exam prep, 70% grade) aur FDE Practicum (governed Vertical SoR banana, 30% grade).

00 — Overview Aur Entry Contract

- Track B = accelerated version (Track A se tez, zyada demanding). 4.5 hrs/week instructor-led + 5-7 hrs guided practice, ~120-150 hrs total quarter mein.
- Entry contract: student commitment + prerequisite skills explicit hain before enroll.

01 — Architect Strand (70% grade)

- Week 1: foundations sprint. Weeks 2-11: CCAR-F blueprint ke 5 domains — Agentic Loop by hand → Agent SDK tools/multi-agent/hooks → Claude Code architecture/Teams/CI → Structured output → Context/reliability.
- 4 required architect projects isi mein bante hain (docs/certifications/ccar-f/projects/ mein scaffold).
- Week 12-13: 6-scenario workshop + 2 full mocks + readiness decision.
- Decision framework: direct execution chhote well-understood changes ke liye; plan mode multi-file/uncertain/multiple-valid-approach kaam ke liye.
- Required behaviour → deterministic control (code/hook/permission gate), sirf prompt guidance nahi kaafi nahi.

02 — FDE Practicum (30% grade)

- P1-P13: ek governed Vertical SoR banata hai — human projection (Fumadocs site) + agent projection (stateless MCP server) same governed corpus se.
- 4 milestones: live human surface (P5), working agent surface (P8), governed KSoR + evaluation (P11), capstone (P13).
- Stateless core: har call independent request/response, koi protocol-level session nahi — server multiple copies mein load-balancer ke peeche chal sake.
- requestState untrusted treat karo, handler re-entrant banao — client-supplied kuch bhi tamper ho sakta hai.

03 — Required Projects Aur Assessment

- Is repo mein scaffold: p4-p5-fumadocs-corpus-to-site, p6-p7-stateless-mcp, p8-agent-surface, p11-proving-behaviour — sab localhost par chalte hain, koi live deployment nahi.
- Assessment split (70%): scenario practice 7%, Projects 1-4 28%, Mock One 7%, Mock Two 21%, trade-off notebook 7%.

04 — Certification Path Aur Sources

- Syllabus khud Panaversity ka document hai (book se bahar); Claude/Anthropic technical facts (stateless MCP) book se cross-check hui hain — context-layer-crash-course confirm karti hai.
- Quiz: 12 self-authored scenario questions (quiz.md), harness-engineering format follow karte hue.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Track B	Accelerated syllabus variant — faster, more demanding than Track A.	Tez, zyada demanding syllabus version (Track A se).
Architect strand	CCAR-F exam-prep track, 70% of grade.	CCAR-F exam-prep track, grade ka 70%.
FDE Practicum	Hands-on Vertical SoR build track, 30% of grade.	Hands-on Vertical SoR banane wala track, grade ka 30%.
Vertical System of Record	Governed corpus with human + agent projections.	Governed corpus jiske human + agent projections hon.
Stateless MCP server	No protocol-level session; any instance can answer any call.	Koi protocol-level session nahi; koi bhi instance koi bhi call answer kar sake.
requestState	Opaque server-generated field round-tripped via client — must stay untrusted.	Server-generated field jo client se round-trip karta hai — untrusted treat karna zaroori.
Plan mode	Used for multi-file / uncertain / multiple-valid-approach work.	Multi-file / uncertain / multiple-valid-approach kaam ke liye use hota hai.
Deterministic control	Code/hook/permission gate enforcing a required behaviour.	Code/hook/permission gate jo required behaviour enforce kare.
Blast radius	Cost of a mistake, not its frequency, decides approval boundary.	Mistake ki cost (na ke frequency) approval boundary decide karti hai.
Milestone evidence	Each milestone has a defined deliverable, e.g. 'stateless MCP search/retrieve/cite'.	Har milestone ka defined deliverable hota hai, jaise 'stateless MCP search/retrieve/cite'.
Three-bin sort	Workflow archaeology: keep / redesign / retire old process elements.	Workflow archaeology: purane process elements ko keep/redesign/retire karna.
Chesterton's Fence	Don't remove a control until you know why it exists.	Jab tak wajah na pata ho, koi control hatao mat.
Useful escalation	States what's verified, names the gap, offers a ready decision.	Verified cheez bataye, gap naam de, ready decision offer kare.
Message Batches	50% cost reduction; no multi-turn tool loop in one request, no latency SLA.	50% cost kam karti hain; ek request mein multi-turn tool loop nahi, latency SLA nahi.
Mock Two	Largest single assessment component (21% of grade).	Sabse bara single assessment component (grade ka 21%).

Viva Prep — Scenario Questions (from quiz.md, 12 total)

#	Scenario Gist	Answer	One-line Reasoning
1	Chhota, isolated, well-understood bug fix ke liye plan mode chalu karna — sahi hai?	**Nahi — direct execution kaafi hai; plan mode multi-file/uncertain kaam ke liye hai.**	Har cheez par plan mode lagana overhead add karta hai bina value ke.
2	Refund control sirf system prompt mein likha hai, code mein nahi — theek hai?	**Nahi — required behaviour ko deterministic control (code/hook/gate) chahiye, sirf prompt nahi.**	Model confused ho sakta hai, context rot ho sakta hai; business invariant prompt wording par depend nahi kar sakti.
5	$50 refund vs $50,000 contract — approval rule kaise differ karegi?	**Blast radius differ karti hai: bara/irreversible action ko explicit human approval chahiye.**	Mistake ki cost (na ke frequency) decide karti hai, 'blast radius, not frequency' principle.
6	MCP server mein in-memory session object 'performance ke liye' banana — stateless core ke against hai?	**Haan — koi protocol-level session nahi honi chahiye, taake server multiple copies mein chal sake.**	In-memory session state guarantee tor deti hai ke koi bhi copy koi bhi request answer kar sake.
8	Thin Vertical SoR ko 'incomplete' samajhna — sahi hai?	**Nahi — thin sirf breadth measure karta hai; ek thin slice bhi poori tarah complete honi chahiye.**	Missing-evidence/wrong-jurisdiction/forbidden-request handle na karna 'unfinished' hai, 'thin' nahi.
10	'I cannot process this request' — acceptable escalation hai?	**Nahi — useful escalation verified cheez bataye, gap naam de, ready decision offer kare.**	Bare 'cannot continue' sab kaam insaan par wapas daal deta hai.

Where This Leads / Sources

Agle qadam: FDE Practicum ke 4 milestones ko [[ecosystem-designing-the-vertical-sor]] ke design method aur [[ksor]] ke governance principles se jorh kar dekho — teeno ek hi 'governed knowledge, two surfaces' pattern ke different angles hain.

docs/ccar-f-fde-track-b/SUMMARY.md, README.md, 00-04 numbered files, quiz.md, 05-test-your-understanding.md, root syllabus .md file.